Impact energy and TNT equivalents

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Learning target: Convert an energy estimate without claiming a complete damage prediction.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 16.*
Same work; the slower student has 1/3 of the faster student's average power.

02. Worked example: explain the model
Same total energy does not fix pressure, coupling, duration, material response or damage.

03. Read and interpret the model
No; both can state energy using different units.

04. Apply the physics
2 tons TNT equivalent.

05. Explain your reasoning
Effects depend on entry, altitude, angle, composition, fragmentation, energy coupling and location.

06. Review: connect an earlier idea *Review: Unit 4, Day 7.*
+350 J; the object gains gravitational potential energy.

07. Exit check
Both quarter.

08. STRETCH: test the model
Energy per area and coupling differ; concentrated transfer can give larger local energy density.

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01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 16.*
P=Fv=150 W; W=Pt=1050 J.

02. Worked example: explain the model
Same total energy does not fix pressure, coupling, duration, material response or damage.

03. Read and interpret the model
No; both can state energy using different units.

04. Apply the physics
Speed 10^4 m/s; K 5×10^{13} J;≈11,950 tons, or 11.95 kilotons.

05. Explain your reasoning
Effects depend on entry, altitude, angle, composition, fragmentation, energy coupling and location.

06. Review: connect an earlier idea *Review: Unit 4, Day 7.*
ΔU=-350 J; gravity does +350 J work.

07. Exit check
Unchanged: total K×2, fraction×0.5.

08. STRETCH: test the model
K=10^9 J; equivalent≈238 kg. Need geometry, materials, coupling, impact angle and other evidence.